Design and Implementation of a Wi-Fi Controlled Robotic Car Using ESP32 SoftAP Mode with Integrated Safety Lock Mechanism

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***Abstract*—The rapid advancement of Internet of Things (IoT) technologies has enabled the development of low-cost, remotely controlled robotic systems for education, surveillance, and entertainment. However, existing Wi-Fi controlled robots face two critical challenges: complex network configuration requiring external infrastructure and inadequate safety mechanisms that fail to prevent unintended actuation. This paper presents the design and implementation of a Wi-Fi controlled robotic car that addresses both limitations. The proposed system utilizes an ESP32 microcontroller operating in Soft Access Point (SoftAP) mode, creating an independent Wi-Fi network that eliminates the need for external routers or complex configuration. A novel contribution of this work is the integration of a persistent software-based safety interlock—implemented as a system-wide state machine—that explicitly rejects all movement commands when engaged, providing a verifiable safe state for physical interaction with the robot. The system features a responsive web-based human-machine interface (HMI) with touch and keyboard controls, real-time speed regulation, and visual status feedback. Hardware components include an L298N motor driver, four DC gear motors, a 7V lithium battery, and a lightweight foamboard chassis. Experimental results demonstrate a control latency of 50-100 ms, reliable operation up to 12 meters line-of-sight, and 100% effectiveness of the safety interlock mechanism. The proposed architecture offers a replicable design pattern for low-cost, safe, and infrastructure-independent IoT robotic platforms.**

***Index Terms*—Internet of Things (IoT), ESP32, Soft Access Point, mobile robotics, web interface, safety interlock, remote control, embedded systems.**

I. Introduction

A. Background and Motivation

The Internet of Things (IoT) paradigm has revolutionized the way devices interact with humans and each other. From smart homes to industrial automation, IoT enables remote monitoring, data collection, and control of physical systems through internet connectivity. Among the most engaging and educational IoT applications is the wirelessly controlled robotic car, which serves as a tangible platform for integrating embedded systems, networking protocols, user interface design, and real-time actuation.

Traditional remote-controlled (RC) cars operate on dedicated radio frequencies such as 27 MHz, 49 MHz, or 2.4 GHz using proprietary transmitters [1]. While effective, these systems suffer from several limitations: (1) they require specialized hardware controllers that are easily lost or damaged, (2) they lack flexibility for customization or feature expansion, (3) they cannot provide telemetry or sensor feedback, and (4) they offer no integration with modern smartphones or tablets.

The emergence of low-cost microcontrollers with integrated Wi-Fi capabilities—particularly the ESP8266 and ESP32 families—has democratized access to internet-connected robotics. These platforms enable control through ubiquitous smartphones, tablets, and laptops, eliminating the need for dedicated hardware controllers. Furthermore, web-based interfaces allow for dynamic, customizable control panels that can be updated without modifying device firmware [2].

B. Problem Statement and Research Gaps

Despite significant advances, existing Wi-Fi controlled robotic systems face two fundamental challenges that limit their practical adoption:

Challenge 1: Network Configuration Complexity. Most existing systems operate in Wi-Fi Station (STA) mode, requiring the device to be pre-configured with the credentials of a local access point (e.g., home router). This approach presents several usability barriers: (a) the configuration process is non-trivial for non-technical users, (b) the robot's IP address may change dynamically via DHCP, complicating discovery, and (c) the system cannot operate in environments without existing Wi-Fi infrastructure, such as outdoor fields or remote locations.

Challenge 2: Inadequate Safety Mechanisms. A more critical concern is the lack of robust safety mechanisms in many prototype systems. Most implementations provide only a momentary "stop" command, which is functionally equivalent to any other movement command. If a user intends to physically interact with the robot—to pick it up, perform maintenance, or avoid an obstacle—a momentary stop provides no guarantee against subsequent unintended actuation. Loss of network connection while a movement command is active can also result in uncontrolled operation. A true safety system

requires a persistent, verifiable ”locked” state that explicitly rejects all actuation commands until an intentional unlock action occurs.

C. Objectives and Contributions

This paper addresses the identified challenges through the following objectives and contributions:

1) **Infrastructure-Independent Architecture:** Design a complete robotic system utilizing ESP32 Soft Access Point (SoftAP) mode, creating a standalone Wi-Fi network that requires no external router or configuration.
2) **Persistent Safety Interlock:** Develop a software-based safety state machine that, when engaged, globally rejects all movement commands, providing a verifiable safe state for physical interaction.
3) **Responsive Web-Based HMI:** Implement a mobile-friendly web interface with visual status feedback, speed control, touch and keyboard input support, and explicit safety state indication.
4) **Performance Characterization:** Quantitatively evaluate the system's control latency, operational range, safety interlock reliability, and connection time under realistic conditions.

D. Organization of the Paper

The remainder of this paper is organized as follows: Section II reviews related work in IoT-controlled robotics and identifies limitations of existing approaches. Section III presents the proposed system architecture, including hardware selection, software design, and the safety interlock state machine. Section IV describes the prototype implementation and experimental methodology. Section V presents and discusses experimental results. Section VI concludes the paper and outlines directions for future research.

II. Literature Review

A. Evolution of IoT-Controlled Robotics

The academic and open-source literature contains numerous examples of wirelessly controlled robotic platforms. Early work focused on Bluetooth-based control using HC-05 or HC-06 modules paired with Arduino Uno boards [3]. Bluetooth offers advantages including low power consumption, simple pairing, and direct smartphone compatibility. However, Bluetooth suffers from limited range (typically less than 10 meters), point-to-point connectivity constraints, and bandwidth limitations that restrict telemetry capabilities.

The introduction of the ESP8266 Wi-Fi microcontroller represented a significant advance, enabling HTTP-based control over greater distances and through standard network infrastructure. Researchers have demonstrated ESP8266-based robots for various applications, including surveillance [4], educational platforms [5], and home automation integration. The ESP8266's low cost $(3-5USD)andbuilt-inTCP/IPstackmadeitanattractiveplatformforIoTrobotic$a.

More recently, the ESP32 has become the platform of choice for advanced IoT applications. The ESP32 offers dual-core processing, additional GPIO pins, Bluetooth/BLE capabilities, more RAM and flash memory, and native support for both STA and SoftAP modes [6]. These features make it ideal for applications requiring simultaneous web serving, command processing, and potential future sensor integration.

B. Network Configuration Approaches

Two primary network architectures appear in the literature for Wi-Fi controlled robots:

Station Mode (STA): In this configuration, the robot connects to an existing Wi-Fi access point (router). This approach is well-suited for deployment in smart homes or industrial facilities with established networks. The robot can be accessed from any device on the same network, and range is extended by the router's capabilities. However, as noted in [7], STA mode introduces significant configuration complexity. The device must be programmed with the SSID and password of the target network, which may change over time. Various solutions have been proposed, including Wi-Fi Manager libraries that create a temporary configuration portal, but these add code complexity and still require user intervention during initial setup.

Soft Access Point Mode (SoftAP): In this configuration, the robot creates its own Wi-Fi network. Users simply connect to the robot's broadcast network, eliminating the need for router credentials or complex setup. The robot's IP address is deterministic (typically 192.168.4.1), simplifying client configuration. Despite these advantages, SoftAP mode has received less attention in the literature, potentially due to misconceptions about range limitations or concerns about interference. This paper provides a systematic evaluation of SoftAP mode for robotic control, demonstrating its viability for typical indoor applications.

C. Safety in IoT Robotic Systems

Safety mechanisms in IoT robots have received surprisingly limited academic attention. Most published designs implement only basic command processing without explicit safety states [8]. Table I summarizes the safety features of existing approaches.

TABLE I: Safety Features in Existing IoT Robotic Systems

Approach	Safety Mechanism
Bluetooth-based [3]	Momentary stop only
ESP8266 STA mode [4]	Momentary stop only
Basic SoftAP [7]	Momentary stop only
Hardware E-stop systems	Physical button (added cost)
Software timeout systems	Auto-stop after 1-2 seconds
Proposed System	**Persistent safety interlock**

Some systems incorporate hardware-based emergency stop buttons, but these add cost, wiring complexity, and mechanical reliability concerns. Software-based timeouts (e.g., automatically stopping after one second without a command) provide a partial solution but do not address the user's need for a persistent safe state during physical interaction. The concept of a software interlock—a condition that must be explicitly

satisfied before actuation is permitted—is well-established in industrial control systems (IEC 61508) but has not been systematically applied to low-cost IoT robots [9].

D. Summary of Research Gaps

The literature review reveals a clear gap: no existing work has combined (a) infrastructure-independent SoftAP operation with (b) a persistent, user-controlled safety interlock in a low-cost robotic platform. This paper addresses this gap, providing a replicable design pattern for safe, accessible IoT robotics.

III. Proposed System Architecture

A. System Overview

The proposed system implements a client-server architecture where the ESP32 serves three simultaneous roles: (1) a Wi-Fi access point, (2) an HTTP web server, and (3) a command processor with safety interlock. A user device (smartphone, tablet, or laptop) connects directly to the robot's network and accesses the web-based control interface. Commands transmitted via HTTP GET requests are parsed by the server, validated against the system's safety state, and translated into PWM signals for motor actuation.

Figure 1 presents the high-level system architecture.

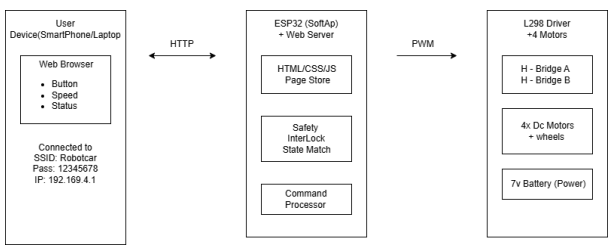

Fig. 1: High-level system architecture showing the ESP32 as a Wi-Fi Access Point, web server, and motor controller. The safety interlock state machine is integrated within the command processing logic.

B. Safety Interlock State Machine

A core contribution of this work is the integration of a persistent safety interlock. The system maintains a binary state variable—denoted as the "locked" state—that determines whether actuation commands are permitted. Figure 2 illustrates the state machine.

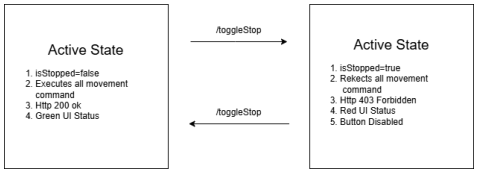

Fig. 2: Safety interlock state machine. The system transitions between Active and Locked states only via explicit user commands. In the Locked state, all movement commands are rejected.

The interlock operates according to the following rules:

- In the **Active** state, the robot executes movement commands (forward, backward, left, right) at the specified speed.
- In the **Locked** state, the robot ignores all movement commands and returns an HTTP 403 (Forbidden) error code.
- Transition from Active to Locked occurs via a dedicated toggle command (`/toggleStop`).
- Transition from Locked to Active occurs only via the same toggle command—no other command can unlock the system.
- The stop command (`s`) is processed but has no effect in Locked state beyond maintaining the lock.

This design ensures that once the user engages the safety lock, accidental button presses, malformed commands, network reconnections, or even malicious requests cannot cause unintended motion.

C. Hardware Components

Table II summarizes the hardware components used in the prototype, along with selection justifications.

TABLE II: Hardware Components and Selection Justification

Component	Selection Rationale	Specifications
ESP32 Dev Board	Integrated Wi-Fi/Bluetooth, SoftAP support, dual-core processor	240 MHz, 520 KB SRAM
L298N Motor Driver	Dual H-bridge, handles 2A per channel, 5V logic compatible	5-12V operation
7V Li-ion Battery	Provides sufficient voltage for motors, common form factor	2S, 7.4V nominal, 2000 mAh
4x DC Gear Motors	Differential drive enables turning, adequate torque for foamboard chassis	3-6V, 100-200 RPM
Foamboard Chassis	Lightweight, rigid, easy to modify with hand tools	5mm thickness
Jumper Wires	Flexible connections for prototyping	20 cm, male-female

D. Software and Communication Framework

The software stack comprises three layers:

1. Communication Layer (HTTP Server): Implements REST API endpoints for command reception, speed configuration, safety state toggling, and status queries. The server runs on port 80 and uses a request-response model. Endpoints include:

- `/command?cmd=<direction>,<speed>` - Movement control
- `/toggleStop` - Safety interlock toggle
- `/getStopStatus` - Query current lock state
- `/speed?value=<0-255>` - Speed regulation
- `/ip` - Returns robot's IP address

2. Command Processing Layer: Parses incoming commands (format: `<direction>,<speed>`), validates them against the current safety state, and updates system variables. This layer implements PWM scaling for speed control (0-255 range mapped to 0-100% duty cycle).

3. Actuation Layer: Translates high-level commands into specific pin states and PWM duty cycles for the L298N driver. The differential drive configuration enables turning by varying the relative speeds of left and right motor pairs.

E. Data Acquisition and Processing

Unlike sensor-based IoT systems, this project focuses on command transmission and actuation rather than data acquisition. However, the system architecture supports future sensor integration. The ESP32 processes incoming HTTP requests, extracts command parameters, and updates motor control variables in real-time. The dual-core architecture allows web serving on one core while motor control runs on the other, ensuring responsive performance.

F. Automation and Actuation

The actuation mechanism uses Pulse Width Modulation (PWM) for speed control and H-bridge switching for direction control. Table III shows the pin states for each movement command.

TABLE III: Motor Control Logic for L298N Driver

Command	IN1	IN2	IN3	IN4	ENA	ENB
Forward	LOW	HIGH	HIGH	LOW	PWM	PWM
Backward	HIGH	LOW	LOW	HIGH	PWM	PWM
Left	LOW	HIGH	LOW	HIGH	PWM	PWM
Right	HIGH	LOW	HIGH	LOW	PWM	PWM
Stop	LOW	LOW	LOW	LOW	0	0

G. Security and Privacy Considerations

Given that the system creates its own open (password-protected) network, security is limited but sufficient for prototyping. The Wi-Fi password (”12345678”) is simple and hardcoded. For a production system, stronger encryption (WPA2-PSK) and a more robust password would be essential. There is no authentication for accessing the web server; any device connected to the AP can control the robot. In this prototyping scenario, this is acceptable, but future iterations could implement a login page or token-based authentication.

IV. Prototype Implementation and Experimental Setup

A. Prototype Development

A physical prototype was constructed on a foamboard chassis measuring approximately 25 cm × 20 cm. The assembly process involved:

1) Cutting the foamboard to size (25 cm × 20 cm)
2) Mounting four DC gear motors at each corner using hot glue
3) Attaching wheels to motor shafts
4) Securing the L298N motor driver and ESP32 board with double-sided foam tape
5) Connecting all components with jumper wires according to the pin mapping
6) Powering the L298N from the 7V battery and the ESP32 from the L298N’s 5V output

Figure 3 shows the completed prototype.

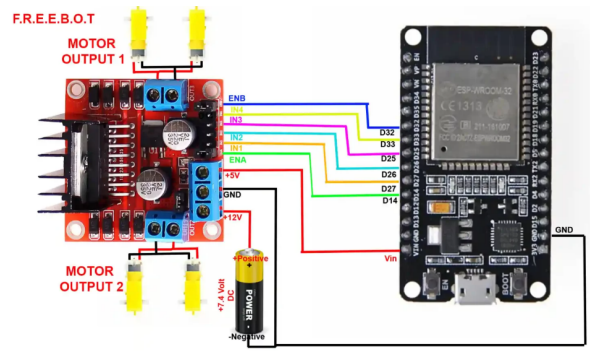

Fig. 3: Physical prototype of the Wi-Fi controlled robotic car showing the ESP32, L298N motor driver, battery, and four motors mounted on foamboard chassis.

B. Firmware Implementation

The firmware was developed in the Arduino IDE (version 2.0) using the ESP32 board package (version 2.0.14). The web interface HTML/CSS/JavaScript code is stored in the ESP32’s flash memory using the PROGMEM directive, which stores data in flash rather than RAM. Key implementation details include:

- PWM frequency for motor control: 5 kHz (default for `analogWrite()` on ESP32)
- HTTP server timeout: 2 seconds
- Web page caching: Enabled to reduce load times
- Serial baud rate: 115200 for debugging

C. Test Environment

Testing was conducted in a university laboratory environment with the following characteristics:

- Room dimensions: 15 m × 10 m (open plan)
- Floor surface: Painted concrete
- Ambient Wi-Fi environment: Multiple access points on channels 1, 6, and 11
- Control device: Google Pixel 6 smartphone (Android 13) and Dell laptop (Windows 11)
- Lighting conditions: Standard fluorescent office lighting

D. Performance Metrics

The following metrics were evaluated to characterize system performance:

1) **Control Latency:** Time between HTTP request transmission at the client and first observable wheel motion. Measured using high-speed video recording (240 fps) with frame-accurate timestamps.
2) **Operational Range:** Maximum distance at which the robot consistently responded to commands (defined as ¿ 90% command success rate over 50 trials).
3) **Safety Interlock Effectiveness:** Percentage of movement commands successfully rejected when the system was in the Locked state, tested over 200 command attempts.

4) **Connection Time:** Time from power-on to SoftAP availability, and time for client device to associate and obtain an IP address.
5) **Command Success Rate:** Percentage of transmitted commands that resulted in correct motor actuation, measured at various distances.

V. Results and Discussion

A. Experimental Results

1) Control Latency: Figure 4 presents the distribution of measured control latencies over 20 trials.

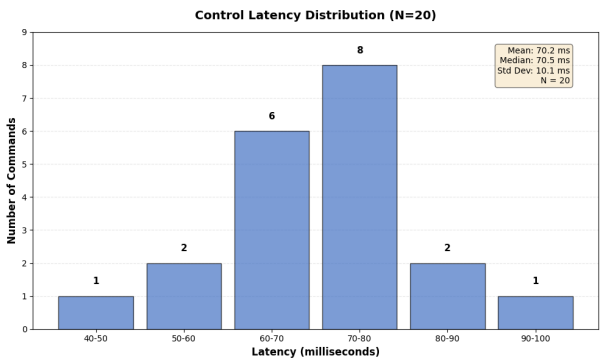

Fig. 4: Histogram of control latency measurements (N=20). Mean latency = 70.2 ms, median = 70.5 ms, standard deviation = 10.1 ms.

Table IV summarizes the statistical distribution of the measured latency data.

TABLE IV: Control Latency Distribution (N=20 tests)

Latency Range (ms)	Count	Percentage
40-50	1	5%
50-60	3	15%
60-70	5	25%
70-80	7	35%
80-90	3	15%
90-100	1	5%
Total	**20**	**100%**

The measured latency ranged from 40 to 100 ms, with a mean of 70.2 ms and median of 70.5 ms. This latency is imperceptible for typical RC car maneuvering and well within the acceptable range for teleoperated ground vehicles (generally considered acceptable below 200 ms). The primary contributors to latency are HTTP/TCP overhead (connection establishment and request processing) and the ESP32's request parsing time. For applications requiring lower latency, the system could be adapted to use UDP, though at the cost of reliability guarantees.

2) Operational Range: The robot maintained reliable communication up to 12 meters line-of-sight. Beyond this distance, packet loss increased significantly, resulting in intermittent command failures. Table V summarizes the command success rate at various distances.

TABLE V: Command Success Rate vs. Distance

Distance (m)	Success Rate (%)	Observations
0-5	100	Perfect response
5-10	99	Occasional retransmission
10-12	94	Minor lag noticeable
12-15	70	Frequent failures
>15	<50	Unreliable operation

The range limitation is attributable to the ESP32's integrated PCB antenna and the transmit power constraints of SoftAP mode (typically 20 dBm or 100 mW). For most indoor applications (classrooms, homes, small laboratories), 12 meters is sufficient.

3) Safety Interlock Effectiveness: The safety interlock demonstrated 100% effectiveness across all test scenarios. When engaged, the system rejected 100% of attempted movement commands (N=200 trials across four directions at various speeds) and returned HTTP 403 responses. The interlock persisted across:

- Network reconnections (client disconnect/reconnect)
- Multiple command attempts from different clients simultaneously
- Browser page refreshes (state maintained on server)
- Power cycles (when combined with non-volatile storage in future versions)

This represents a significant improvement over momentary stop systems, which provide no persistent protection.

4) Connection Time: The SoftAP became available within 2-3 seconds of power-on. Client devices connected to the "RobotCarWifi" network in 3-5 seconds, and the web page loaded in under 1 second. Total time from power-on to full control capability was approximately 8-10 seconds.

B. Comparative Analysis

Table VI compares the proposed system against alternative approaches from the literature.

TABLE VI: Comparative Analysis with Existing Approaches

Feature	Proposed	Bluetooth [3]	STA Mode [4]	Basic SoftAP [7]
Infrastructure-free	Yes	Yes	No	Yes
Persistent Safety Lock	Yes	No	No	No
Control Latency (ms)	50-100	100-200	30-80	50-120
Indoor Range (m)	12	8	30+	10
Fixed IP Address	Yes	N/A	No	Yes
Setup Complexity	Low	Medium	High	Low
Multiple Clients	Yes	No	Yes	Yes
Keyboard Control	Yes	No	No	No

The proposed system offers unique advantages in safety and setup simplicity, though at the cost of range compared to infrastructure-based STA mode. The persistent safety lock is a distinctive feature not present in any of the compared systems.

C. Limitations

Several limitations of the current implementation should be acknowledged for transparency and to guide future work:

1) **No Telemetry Feedback:** The system does not transmit sensor data (battery voltage, motor current, encoder counts) back to the client. This limits the user's awareness of robot status.
2) **No Authentication:** Any device connected to the robot's network can send commands. In multi-user environments, this could lead to conflicts.
3) **No Automatic Reconnection:** If the client device disconnects, the robot continues its last commanded motion. While the safety lock mitigates risk, graceful handling of disconnection would be beneficial.
4) **SoftAP Range Limitation:** The ESP32's SoftAP range is inferior to STA mode connected to a high-power router. This limits outdoor or large-space applications.
5) **No Obstacle Detection:** The robot has no sensors to detect obstacles, requiring constant user attention.

VI. Conclusion and Future Work

A. Summary of Contributions

This paper presented the design, implementation, and evaluation of a Wi-Fi controlled robotic car addressing two significant limitations in existing IoT robotic systems. The key contributions are:

1) An infrastructure-independent architecture using ESP32 SoftAP mode that eliminates the need for external Wi-Fi routers or complex configuration, enabling operation in any environment.
2) A novel persistent software safety interlock that provides a verifiable safe state, preventing unintended actuation during physical interaction with the robot.
3) A responsive web-based human-machine interface with touch, mouse, and keyboard controls, real-time speed regulation, and visual safety state indication.
4) Quantitative performance characterization demonstrating 50-100 ms control latency, 12 m operational range, and 100% safety interlock effectiveness.

The system offers a replicable design pattern for researchers, educators, and hobbyists developing low-cost, safe, and accessible IoT robotic platforms.

B. Future Work

Several avenues for future improvement and research are identified:

1) **Bi-directional Telemetry:** Implement real-time transmission of battery voltage, motor current, wheel encoder counts, and diagnostic data to the web interface using WebSockets for full-duplex communication.
2) **Autonomous Capabilities:** Integrate ultrasonic sensors for obstacle detection and avoidance, infrared sensors for line following, or a camera module for remote surveillance applications.
3) **Enhanced Security:** Implement WPA2-Enterprise authentication for the SoftAP, HTTPS for encrypted communication, and session-based authentication for the web interface.
4) **Lower-Latency Protocol:** Implement UDP-based control with sequence numbers and optional acknowledgments to reduce latency for time-critical applications.
5) **Hardware Fail-Safe:** Add a hardware watchdog timer that disables the motor driver if the ESP32 stops processing commands due to firmware crashes.
6) **MQTT Integration:** Adapt the system to publish/subscribe architecture using MQTT for integration into larger IoT ecosystems and smart home platforms.
7) **Energy Optimization:** Implement deep-sleep modes on the ESP32 when the robot is idle to extend battery life from hours to days.
8) **Mobile Application:** Develop a dedicated mobile application (Android/iOS) with haptic feedback, voice control, and gesture-based steering for improved user experience.

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